A GROUP 6 DESIGN THINKING INITIATIVE

TRY BEFORE YOU BUY™: SOLVING THE RETAIL FIT CRISIS THROUGH AI VIRTUAL TRY-ON

THE RETAIL FIT CRISIS

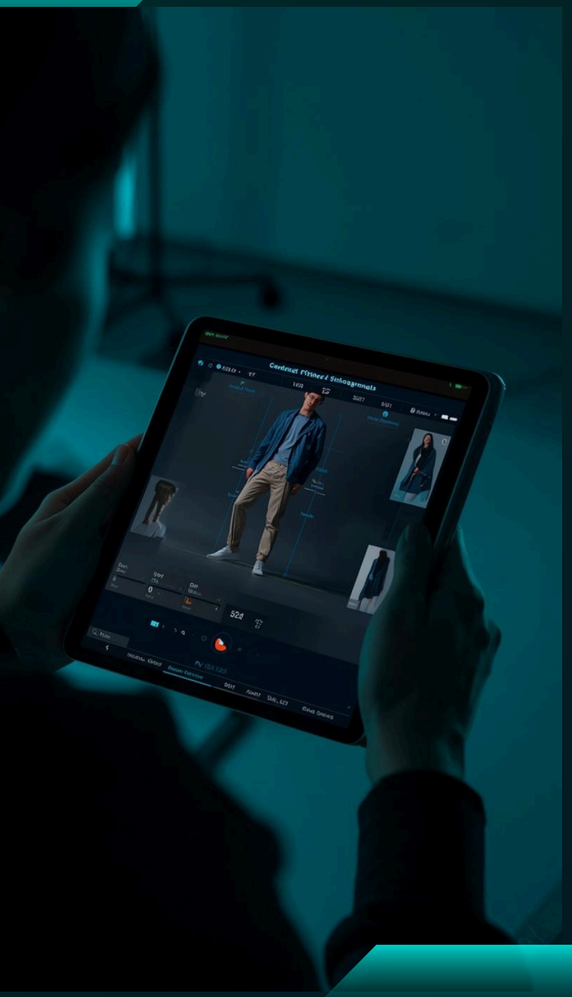
Online apparel shopping faces a massive challenge: a 30% average return rate costing retailers $60 billion annually. The root cause? Fit and size uncertainty—shoppers simply cannot try on clothes before purchasing. Without the tactile experience of a fitting room, customers order multiple sizes hoping one fits, leading to wasteful returns and frustrated experiences.

Local boutiques offer personalized service and try-on experiences, but lack the inventory variety and convenience of online shopping. How do we bridge this gap?

CORE INSIGHT – THE EMOTIONAL DIMENSION OF RETURNS

Traditional analytics track what happens — items returned, sizes exchanged, refunds processed. But they miss the deeper story. Our ethnographic interviews and surveys revealed that returns are fundamentally driven by emotion: hesitation at checkout, guilt over keeping items that don't feel right, and distrust in sizing accuracy.

Design Thinking allowed us to uncover why confidence is lacking in the first place. Shoppers expressed feelings like "I never know if it will actually fit me" and "I feel bad returning things, but I can't afford to keep mistakes." This human-centered approach shifts focus from processing returns to preventing the emotional uncertainty that causes them.



PROBLEM REFRAMING

Design Thinking shifts our focus from reactive solutions to proactive prevention. Instead of optimizing the returns process, we ask: what if returns never needed to happen? This reframe transforms how we approach the retail fit crisis.

OLD

Traditional Question:
"How do we process retail returns faster and cheaper?"
Focus: Speed & cost reduction
Approach: Reactive logistics
Outcome: Still losing $60B annually

NEW

Reframed Question:
"How might we eliminate fit uncertainty and build absolute user trust?"
Focus: Prevention & confidence
Approach: Proactive design
Outcome: Returns don't happen

DIVERSE USER NEEDS

01

Online Shopper & Boutique Customer: Seeks sizing confidence to avoid wasteful returns. Wants quick, reliable try-on experience without the hassle of physical fitting rooms or shipping items back.

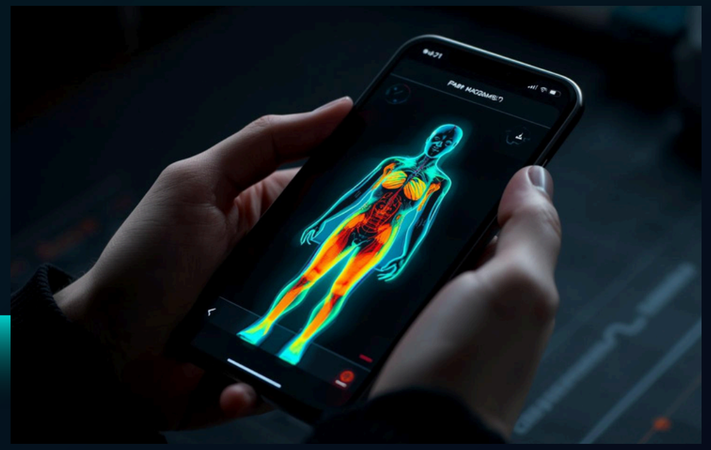
02

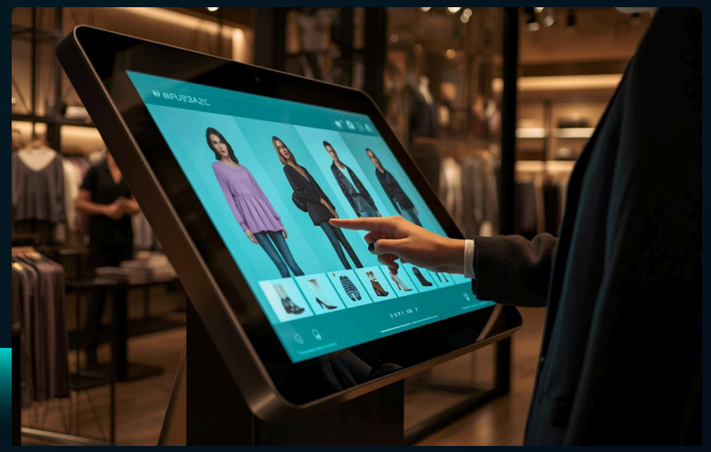
Boutique Owner: Needs to reduce operational costs and minimize unsold inventory. Looking for solutions that enhance customer experience while improving business efficiency.

03

Tech-Savvy & Privacy-Conscious Users: Demands accuracy and transparency in AI recommendations. Requires secure, private data handling with full control over personal information.

BRAINSTORMED IDEAS



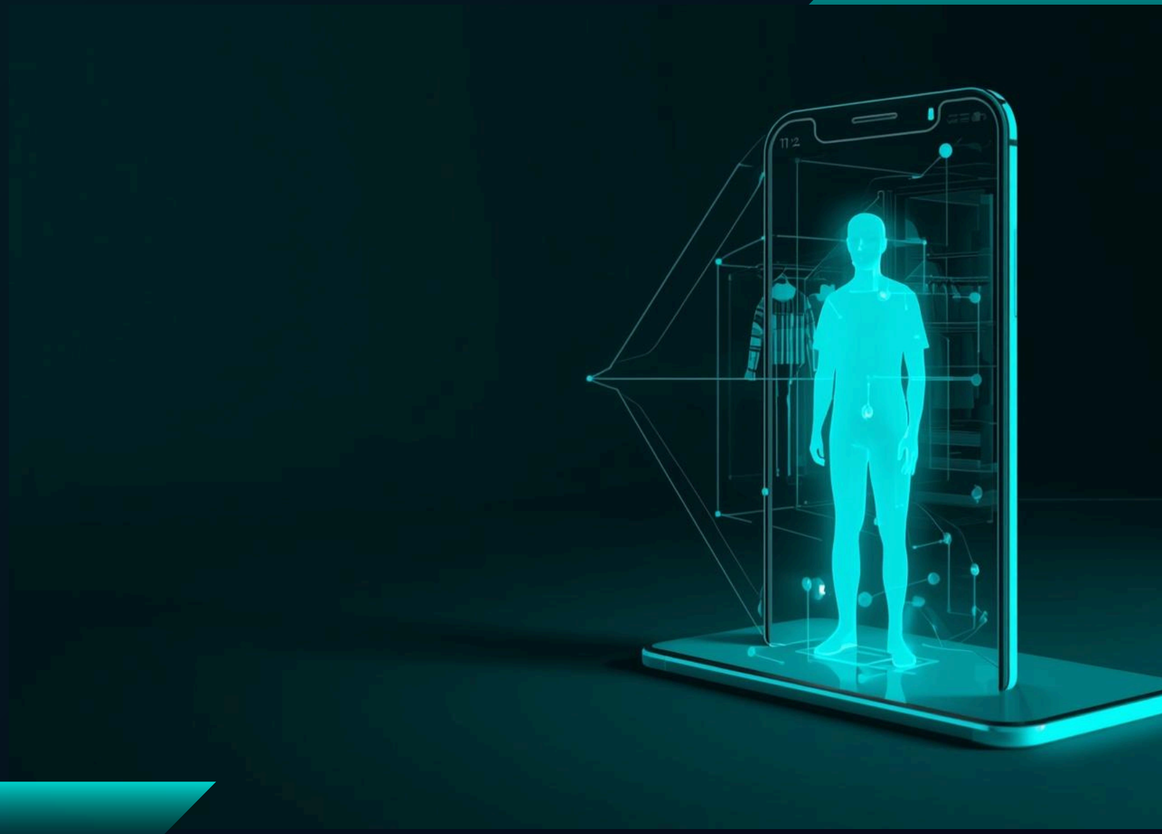


• Mobile 3D Body Scanner – Full body scan in under 1 minute using only a smartphone
• In-Store Endless Aisle Kiosks – Digital try-on stations for boutiques
• Zero-Retention Data Vault – Privacy-first local rendering with no photo storage
• Fit Heatmaps – Visual display of garment tightness and looseness
• Universal Sizing Passport – Cross-retailer sizing profile
• Shared Virtual Cart – Social shopping for friend feedback

KEY INNOVATION

The Phygital Try-On Ecosystem represents our core innovation—a revolutionary approach that seamlessly bridges digital convenience with physical precision. At its heart lies a decentralized, privacy-first 3D digital avatar that empowers users across both online and local boutique environments. This unified technology ensures consistent, accurate fit visualization whether shopping from home or visiting a neighborhood store.

One avatar. Two worlds. Zero compromise on privacy or precision—delivering the confidence of in-store try-on with the convenience of digital shopping.



USER JOURNEY

01

Capture
Using just your smartphone, take a few photos or a short video of yourself. No special equipment needed—the process takes under 60 seconds and works with any modern mobile device.

02

Generate
AI photogrammetry instantly creates your personal 3D avatar with 80+ precise body measurements. Your digital twin is rendered locally with zero data retention for complete privacy.

03

Browse, Assess & Purchase
Visualize clothing on your avatar with fit heatmaps highlighting tension points. Make confident purchase decisions without guesswork—no more returns due to sizing uncertainty.

IDEA VALIDATION & IMPACT

Our research validates the AI Virtual Try-On solution through comprehensive user surveys, technical feasibility studies, and business impact analysis. The data demonstrates strong consumer trust, proven technological accuracy, and significant business benefits that extend to environmental sustainability.

01

User Trust & Tech Viability
70% of surveyed users trust AI fit recommendations. Virtual try-on technology achieves 96-97% measurement accuracy in under 45 seconds, validated by platforms like 3DLOOK and TrueToForm.

02

Business & Environmental Impact
AI-powered virtual try-ons reduce return rates by 40%, cutting the $60B annual return cost. This reduces need for discounts, builds customer loyalty, and lowers environmental footprint by minimizing returns and packaging waste.

NEXT STEPS

- Prototype Privacy UI; test with privacy-conscious users
- Launch local boutique pilot with iPad kiosk
- Develop API mapping AI measurements to brand size charts
- Contact Group 6 for collaboration opportunities

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THANK YOU

Thank you for your attention! We welcome any questions about our AI Virtual Try-On solution and the "Try Before You Buy" initiative. Let's transform the retail experience together.
— Group 6 Design Thinking Initiative